



Physiological Stressors

Exercise-induced vomiting and near-syncope (fainting) during endurance events like a 10-kilometer run are common but complex physiological responses. This phenomenon often results from a combination of intense physical exertion, gastrointestinal distress, and inadequate physiological adaptation to extreme stress. Each factor represents an intersection of the body's cardiovascular, respiratory, neurological, and metabolic systems, which become strained in response to prolonged exercise. Below, we examine the primary reasons why a person may experience nausea, vomiting, and near-syncope while running.

1. Inadequate Cardiovascular Response

During prolonged aerobic exercise, the cardiovascular system undergoes significant demands to deliver oxygen to active muscles. In a 10-kilometer run, cardiac output increases, while blood is shunted away from the GI tract to meet the metabolic demands of skeletal muscles. As blood flow to non-essential organs is reduced, the digestive system receives minimal circulation, making it difficult to absorb fluids and nutrients and

creating a feeling of gastrointestinal discomfort. This shift in blood flow can cause the stomach to become more sensitive, increasing the likelihood of nausea and, eventually, vomiting.

25 Furthermore, in some individuals, prolonged cardiovascular stress can lead to a brief, acute drop in blood pressure as they approach the limits of their endurance, contributing to a sensation of near-fainting.

2. Heat Stress and Thermoregulation

30 Running generates substantial body heat, which the body attempts to dissipate through sweating and increased skin blood flow. However, thermoregulation demands additional cardiovascular resources. When combined with high temperatures or intense effort, core body temperature can rise rapidly, leading to heat
35 stress and dehydration. This thermoregulatory stress may impair brain function temporarily, as overheated neurons can struggle to maintain their activity levels, producing symptoms of dizziness or near-syncope. Dehydration compounds these effects, further lowering blood volume, hindering circulation, and amplifying the
40 risk of vomiting and fainting.

3. Gastrointestinal Distress and Abdominal Pressure

Running exerts continuous mechanical stress on the abdominal organs. This “jostling” effect agitates the stomach, particularly if there is food or fluid present. During intense exercise, the lower
45 esophageal sphincter (LES), which normally prevents gastric contents from rising, can weaken, especially under the pressure changes induced by rapid breathing and the pounding motion of running. As a result, the body may reflexively induce vomiting to relieve the discomfort caused by gastric contents pressing against
50 the LES.

Additionally, exercise can activate the autonomic nervous system in a way that exacerbates gastric emptying. High-intensity running may delay gastric emptying, leaving undigested food and fluids in the stomach longer than usual, which leads to nausea and, ultimately, vomiting.

4. Metabolic Imbalance and Hypoglycemia

Energy demand is elevated during a 10-kilometer run, and without proper fueling, runners may experience hypoglycemia (low blood sugar), resulting in weakness, dizziness, and nausea.

Hypoglycemia triggers the release of stress hormones like cortisol and epinephrine, which work to increase blood sugar levels by mobilizing stored glucose from the liver. However, these hormones also stimulate the brain's vomiting center in the medulla, leading to nausea or vomiting. As glucose levels drop, the brain becomes deprived of its primary energy source, which can cause lightheadedness and near-fainting. This condition is often exacerbated when runners do not consume adequate carbohydrates before their run or if they run on an empty stomach.

5. Central Nervous System Fatigue and Psychological Stress

Endurance running imposes considerable demands on the central nervous system (CNS), particularly when pushing past physical limits. As the CNS becomes fatigued, it struggles to coordinate the physiological processes involved in maintaining posture, balance, and gastrointestinal regulation, potentially triggering dizziness or fainting. Moreover, psychological stress—whether from performance anxiety or the anticipation of physical exhaustion—can heighten the activation of the sympathetic nervous system. This heightened sympathetic activation disrupts the balance of the

80 autonomic nervous system and can aggravate nausea and the
feeling of near-syncope. Psychological stressors also impact the
perception of fatigue, with a cascade of neurochemical changes
that may worsen the sense of physical discomfort.

6. Hydration and Electrolyte Imbalances

85 Fluid and electrolyte balance is critical during prolonged exercise,
and improper hydration practices can lead to hyponatremia (low
sodium levels) or dehydration. Both conditions disrupt cellular
fluid balance, which can impair cognitive function, cause
disorientation, and lead to vomiting. When dehydration or
90 electrolyte imbalances reach severe levels, they can cause muscle
cramps, abdominal discomfort, and impair the brain's ability to
maintain consciousness, increasing the risk of fainting. Sodium
depletion in particular can affect neural communication and may
cause symptoms of nausea and confusion, further increasing the
95 likelihood of vomiting and near-syncope.

Conclusion

Exercise-induced vomiting and near-syncope during endurance
running represent a combination of mechanical, cardiovascular,
and neurological factors. As the body struggles to meet the intense
100 physiological demands of a prolonged run, multiple systems may
become overstressed, leading to a cascade of symptoms that can
include nausea, vomiting, and a sensation of fainting.
Understanding these responses highlights the complex interplay
between the body's need for oxygen, temperature regulation,
105 metabolic stability, and neural coordination, underscoring the
importance of proper preparation, training, and pacing to prevent
these outcomes during endurance events.